Azure PaaS & Azure Cloud

Part 1 (Azure Fundamentals)

[Interview Q&A .NET Developer (23 modules)]

Question 1: What are IaaS, PaaS, and SaaS? Explain with Azure examples.

Interviewer

What is the difference between IaaS, PaaS, and SaaS? Which Azure services belong to each category?

Answer

Cloud services are divided into three categories.

Feature	IaaS	PaaS	SaaS
Infrastructure managed by	Customer	Azure	Vendor
OS management	Customer	Azure	Vendor
Application deployment	Customer	Customer	Vendor
Maintenance	Customer	Azure	Vendor

IaaS (Infrastructure as a Service)

Azure provides infrastructure only.

Examples

- Azure Virtual Machines
- Virtual Machine Scale Sets
- Virtual Networks
- Load Balancer
- Managed Disks

Architecture

Application

↓

Operating System

↓

Virtual Machine

↓

Azure Infrastructure

You manage

- Windows/Linux
- IIS
- Runtime
- Application
- Security patches (unless automated)

PaaS (Platform as a Service)

Azure manages infrastructure and platform.

You deploy only the application.

Examples

- Azure App Service
- Azure Functions
- Azure SQL Database
- Azure Cosmos DB
- Azure Service Bus

Architecture

Application

↓

Azure Platform

↓

Azure Infrastructure

Advantages

✓ No OS maintenance

✓ Auto Scaling

✓ High Availability

✓ Faster deployment

SaaS (Software as a Service)

Complete software.

Examples

- Microsoft 365
- Outlook Online
- Dynamics 365

Users simply use the application.

Interview Question

Which model is preferred for modern .NET applications?

Senior Answer

PaaS, because it reduces operational overhead, provides built-in scaling and monitoring, and allows development teams to focus on delivering business functionality rather than managing infrastructure.

Question 2

Explain Azure App Service

One of the most frequently asked Azure questions.

Answer

Azure App Service is a fully managed PaaS offering for hosting:

- ASP.NET Core APIs

- MVC applications
- Web applications
- Node.js
- Java
- Python
- PHP

Architecture

Internet

↓

Azure Front End

↓

App Service

↓

ASP.NET Core API

↓

Azure SQL

Features

✓ Auto Scaling

✓ Deployment Slots

✓ Custom Domains

✓ SSL Certificates

✓ Authentication

✓ Managed Identity

✓ Logging

✓ Application Insights

Deployment Methods

- Azure DevOps
- GitHub Actions
- ZIP Deploy
- Azure CLI
- Visual Studio Publish

Interview Question

Why choose App Service over Virtual Machines?

Senior Answer

App Service	**Virtual Machine**
PaaS	IaaS
Azure manages OS	You manage OS
Built-in scaling	Manual configuration
Faster deployment	More administration
Lower operational overhead	Higher operational overhead

Real-Time Example

Insurance API

Internet

↓

Azure Front Door

↓

App Service

↓

Azure SQL

↓

Key Vault

Question 3

Explain Azure Functions

Extremely common.

Answer

Azure Functions is Microsoft's **serverless compute** service.

You write code.

Azure manages:

- Servers
- Scaling
- Runtime
- Infrastructure

Trigger Types

Trigger	**Example**
HTTP	REST API
Timer	Daily job
Blob	File upload

Trigger	Example
Queue	Background processing
Service Bus	Message processing
Event Grid	Event-driven architecture
Cosmos DB	Change Feed processing

Example

```
[FunctionName("GetEmployee")]
public async Task<IActionResult> Run(
   [HttpTrigger] HttpRequest req)
{
   return new OkObjectResult("Hello");
}
```

Real-Time Scenario

Blob Uploaded

↓

Blob Trigger

↓

Resize Image

↓

Save Thumbnail

Interview Question

When should you use Azure Functions?

Use cases

- Scheduled jobs
- File processing
- Queue processing
- Background tasks

- Lightweight APIs
- Event-driven workflows

Avoid using Functions for long-running, stateful business processes unless you use Durable Functions or another orchestration mechanism.

Question 4

Explain Azure Storage Account

Very frequently asked.

Azure Storage provides several services.

Service	Purpose
Blob Storage	Files and images
Queue Storage	Messaging
Table Storage	NoSQL key-value data
File Storage	SMB file shares

Blob Storage

Stores

- PDFs
- Images
- Videos
- Documents
- Backups

Example

Container

↓

employeephotos

↓

photo.jpg

Queue Storage

Example

Upload File

↓

Queue Message

↓

Azure Function

↓

Process File

Table Storage

Stores NoSQL entities.

Suitable for

- Logging
- Configuration
- Lightweight metadata

Azure Files

Shared network storage using SMB.

Interview Question

Difference between Blob and Azure Files?

Blob	Azure Files
Object Storage	File Share
HTTP/HTTPS access	SMB protocol
Images, videos, backups	Shared files for applications

Question 5

Explain Azure Service Bus

One of the top Azure interview questions.

Answer

Azure Service Bus is an enterprise messaging service that enables reliable communication between distributed applications.

Why use Service Bus?

Instead of

Application A

↓

Application B

Direct communication,

Use

Application A

↓

Service Bus

↓

Application B

Applications become loosely coupled.

Components

Queue

One sender

↓

One receiver

Topic

One sender

↓

Multiple subscribers

Real-Time Example

Insurance Claim

Customer

↓

Submit Claim

↓

Service Bus Queue

↓

Claim Processing

↓

Email Service

↓

Audit Service

Advantages

✓ Reliable Messaging

✓ Retry

✓ Dead Letter Queue

✓ Duplicate Detection

✓ FIFO support (Sessions)

Interview Question

Difference between Queue and Topic?

Queue	**Topic**
One consumer	Multiple subscribers
Point-to-point	Publish-subscribe

Question 6

Explain Azure Event Grid

Event Grid

↓

Event Routing Service

Example

Blob Uploaded

↓

Event Grid

↓

Azure Function

↓

Resize Image

Use Cases

- Blob events
- Resource events
- Custom application events

Interview Question

Difference between Service Bus and Event Grid?

Service Bus	**Event Grid**
Messaging	Event routing
Reliable delivery	Event notifications
Supports queues/topics	Publishes events to subscribers

Question 7

Explain Azure Event Hub

Event Hub

↓

Big Data Streaming

Examples

- IoT
- Telemetry
- Application Logs
- Clickstream Data

Real-Time

1 Million Devices

↓

Event Hub

↓

Azure Stream Analytics

↓

Power BI

Interview Question

Difference between Event Hub and Service Bus?

Senior Answer

Use **Event Hub** for high-throughput telemetry and streaming data.

Use **Service Bus** for enterprise messaging, workflows, and reliable business transactions.

Question 8

Explain Azure SQL Database

Azure SQL Database is a fully managed relational database service.

Advantages

✓ Automatic Backups

✓ High Availability

✓ Geo-Replication

✓ Patching Managed by Azure

✓ Elastic Scaling

Interview Question

Difference between SQL Server and Azure SQL Database?

SQL Server	**Azure SQL Database**
Customer manages server	Azure manages infrastructure
Manual patching	Automatic patching
IaaS or on-premises	PaaS

Question 9

Explain Azure Cosmos DB

Very important.

Cosmos DB

↓

Globally Distributed NoSQL Database

Supports

- SQL API
- MongoDB API
- Cassandra API
- Gremlin API
- Table API

Features

✓ Global Distribution

✓ Multi-region replication

✓ Low latency

✓ Automatic indexing

✓ Multiple consistency levels

Interview Question

Why choose Cosmos DB over SQL Server?

Senior Answer

Cosmos DB is ideal for globally distributed applications requiring flexible schemas, low latency, and horizontal scalability. SQL Server is better suited for strongly relational workloads with complex joins and transactional consistency.

Question 10

Explain Azure Key Vault

One of the most frequently asked Azure questions.

Azure Key Vault securely stores:

- Secrets
- Certificates
- Encryption Keys

Instead of

appsettings.json

↓

Password

Store

↓

Azure Key Vault

Example

Application

↓

Managed Identity

↓

Key Vault

↓

Connection String

Advantages

✓ Secure

✓ Centralized

✓ Auditing

✓ Rotation Support

✓ No hardcoded secrets

Interview Question

How does your application access Key Vault without storing credentials?

Senior Answer

Use **Managed Identity**.

The application authenticates to Azure automatically, receives an Azure AD token, and accesses Key Vault based on assigned RBAC roles or access policies (depending on the Key Vault permission model).

⭐ **Senior Interview Scenario**

Interviewer

You deployed an ASP.NET Core API to Azure App Service.

The application cannot read a secret from Azure Key Vault.

How do you troubleshoot?

Strong Answer

I would verify:

1. The App Service has a **System-Assigned** or **User-Assigned Managed Identity** enabled.
2. The Managed Identity has the required **Key Vault permissions** (Azure RBAC or Access Policy, depending on configuration).
3. The Key Vault firewall and networking settings allow access (Private Endpoint, trusted services, or public access as appropriate).
4. The correct Key Vault URI is configured.
5. The application is using DefaultAzureCredential or the intended credential chain correctly.
6. Application logs and Azure Monitor for authentication or authorization failures.
7. Azure AD sign-in logs if authentication appears to be failing.

Module 5 Progress

We've covered:

- IaaS vs PaaS vs SaaS
- Azure App Service
- Azure Functions
- Azure Storage Account
- Azure Service Bus
- Azure Event Grid
- Azure Event Hub
- Azure SQL Database
- Azure Cosmos DB
- Azure Key Vault

Next Part (Module 5 – Part 2)

We'll cover advanced Azure topics that are heavily tested in senior interviews:

- Managed Identity (System vs User Assigned)

- Azure RBAC
- Azure API Management (APIM)
- Azure Application Insights
- Azure Monitor
- Azure Storage Authentication
- Azure Networking (VNet, NSG, Private Endpoint)
- Azure Front Door
- Azure Load Balancer
- Deployment Slots
- Scaling
- Real-world production scenarios

Based on our previous conversations, this section will be especially valuable because you've worked with **Managed Identity, Cosmos DB RBAC, Azure Storage, Bicep, VMSS, and Network Security Perimeters**. I'll include interview scenarios similar to the production issues you've encountered.

Excellent. This is the **most important Azure section** for your profile.

Based on our previous discussions, you've worked with:

- Azure Managed Identity
- Azure Cosmos DB
- Azure Storage
- Azure Key Vault
- Azure App Service
- VMSS
- Bicep
- Azure DevOps
- Azure RBAC
- Network Security Perimeter (NSP)

These are exactly the areas interviewers focus on for **Senior Azure/.NET Developers (8–12 years)**.

Module 5 – Azure PaaS & Azure Cloud

Part 2 (Advanced Azure)

Question 11: Explain Managed Identity

Interviewer

What is Managed Identity? Why is it better than storing secrets?

Answer

Managed Identity is an Azure feature that allows Azure resources to authenticate to Azure services **without storing credentials** in the application.

Instead of

Application

↓

Username

Password

Use

Application

↓

Managed Identity

↓

Azure AD Token

↓

Azure Resource

Benefits

✓ No passwords

✓ No secret rotation

✓ Secure authentication

✓ Azure AD integration

✓ Automatic token management

Interview Tip

Say this:

Managed Identity eliminates the need to store secrets in configuration files or source code by using Azure AD-issued tokens.

Senior interviewers love this answer.

Question 12

Difference between System Assigned and User Assigned Managed Identity

Very Frequently Asked

System Assigned

App Service

↓

Managed Identity

↓

Deleted with App

Characteristics

- One Identity
- One Resource
- Lifecycle tied to resource

User Assigned

User Assigned Identity

↓

App Service

↓

Azure Function

↓

VMSS

One identity

↓

Multiple resources

Comparison

System Assigned	User Assigned
One resource	Multiple resources
Deleted with resource	Independent lifecycle
Simpler	Reusable

Interview Question

When would you use User Assigned Identity?

Strong Answer

When multiple Azure resources need to share the same identity or when the identity lifecycle should be independent of any single resource.

Example:

Azure Function

↓

App Service

↓

VMSS

↓

Same User Assigned Identity

↓

Key Vault

Question 13

Explain Azure RBAC

RBAC

stands for

Role-Based Access Control.

Azure controls

Who

Can perform

Which Action

On

Which Resource

Example

Developer

↓

Storage Blob Data Contributor

↓

Storage Account

Built-in Roles

- Owner
- Contributor
- Reader
- User Access Administrator
- Storage Blob Data Contributor
- Key Vault Secrets User
- Cosmos DB Built-in Data Contributor

Interview Question

Difference between

Contributor

and

Owner

Contributor	**Owner**
Manage resources	Manage resources + permissions
Cannot grant access	Can assign roles

Real-Time Example

App Service

↓

Managed Identity

↓

Role

↓

Storage Blob Data Contributor

↓

Blob Storage

Question 14

Explain Azure API Management (APIM)

Very Popular

APIM

acts as

API Gateway

Architecture

Clients

↓

Azure API Management

↓

Employee API

↓

Order API

↓

Policy API

Features

✓ Authentication

✓ Rate Limiting

✓ Caching

✓ Logging

✓ Request Transformation

✓ Response Transformation

✓ API Versioning

✓ Developer Portal

Interview Question

Why use APIM?

Strong Answer

To centralize cross-cutting concerns such as security, throttling, monitoring, transformation, and versioning without modifying backend services.

Question 15

Explain Azure Application Insights

Very Frequently Asked

Application Insights

Monitors

- Requests
- Exceptions
- Dependencies
- Performance

- Availability

Architecture

Application

↓

Application Insights

↓

Azure Monitor

Captures

✓ Response Time

✓ SQL Calls

✓ REST API Calls

✓ Exceptions

✓ User Activity

Interview Question

How do you troubleshoot a slow API?

Expected Answer

Check

Application Insights

↓

Requests

↓

Dependencies

↓

Exceptions

↓

SQL Calls

↓

External APIs

Question 16

Explain Azure Monitor

Azure Monitor

collects

- Metrics
- Logs
- Alerts

Across Azure resources.

Architecture

App Service

↓

Azure Monitor

↓

Alerts

↓

Email

↓

Teams

↓

Webhook

Interview Question

Difference between

Azure Monitor

and

Application Insights

Azure Monitor	**Application Insights**
Platform monitoring	Application monitoring
VMs, Storage, Networking	Application code, requests, dependencies

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Question 17

Explain Storage Authentication

Very Common

Azure Storage supports

- Shared Key
- SAS Token
- Azure AD (Managed Identity)

Shared Key

Uses Storage Account Key.

Not recommended for new applications.

SAS

Provides temporary access.

Example

Valid

↓

30 Minutes

Managed Identity

Best Practice

Application

↓

Managed Identity

↓

Storage Blob

Interview Question

Which authentication method is preferred?

Answer

Managed Identity with Azure RBAC because it avoids shared secrets and follows the principle of least privilege.

Question 18

Explain Private Endpoint

Very Important

Without Private Endpoint

Application

↓

Internet

↓

Storage

With Private Endpoint

Application

↓

VNet

↓

Private Endpoint

↓

Storage

Advantages

✓ No Public Internet

✓ More Secure

✓ Private IP

Interview Question

Why use Private Endpoint?

To ensure Azure services are accessed privately over the Azure network instead of the public internet.

Question 19

Explain Azure Front Door

Azure Front Door

↓

Global Load Balancer

Features

✓ SSL Offloading

✓ WAF

✓ Global Routing

✓ Health Probes

✓ CDN Integration

Architecture

Users

↓

Azure Front Door

↓

West US

↓

East US

↓

Europe

Interview Question

Difference between

Azure Front Door

and

Load Balancer

Azure Front Door	**Azure Load Balancer**
Global Layer 7	Regional Layer 4
HTTP/HTTPS	TCP/UDP
Web applications	Network traffic

Question 20

Explain Deployment Slots

Very Frequently Asked

Slots

Production

↓

Staging

↓

Testing

Deploy

↓

Staging

↓

Test

↓

Swap

↓

Production

Advantages

✓ Zero Downtime

✓ Easy Rollback

✓ Safe Deployment

Interview Question

What happens during Swap?

Azure exchanges the production and staging slots (subject to slot-specific settings), allowing the new version to become live with minimal downtime.

⭐ **Scenario-Based Questions**

Scenario 1

Interviewer

Your App Service cannot access Azure Blob Storage using Managed Identity.

How do you troubleshoot?

Strong Senior Answer

I would verify the following:

1. Managed Identity is enabled on the App Service.
2. The correct identity (System or User Assigned) is being used.
3. The identity has the **Storage Blob Data Contributor** role (or appropriate role).

4. Role assignment has propagated.
5. The storage account networking allows access (Private Endpoint, firewall, trusted services as applicable).
6. The application uses DefaultAzureCredential or the correct Azure Identity credential.
7. Application logs and Azure Activity Logs for authorization failures.

Scenario 2

Interviewer

Application suddenly cannot connect to Azure SQL Database.

How do you troubleshoot?

Strong Answer

Check:

- Azure SQL firewall rules
- Managed Identity or authentication configuration
- Connection string
- Azure AD permissions (if using Azure AD authentication)
- Network configuration (VNet, Private Endpoint, DNS)
- SQL availability and metrics
- Connection pool exhaustion
- Recent deployments or configuration changes

Scenario 3

Interviewer

Blob upload is returning HTTP 403.

What could be the reasons?

Expected Senior Answer

Possible causes include:

- Missing RBAC role assignment
- Expired or invalid SAS token

- Managed Identity lacks permissions
- Storage firewall or networking restrictions
- Private Endpoint/DNS misconfiguration
- Incorrect storage account or container
- Clock skew affecting authentication tokens (less common)

Top 15 Rapid-Fire Azure Questions

Q1. What is Managed Identity?

A secure Azure identity for authenticating to Azure services without storing credentials.

Q2. System vs User Assigned Identity?

System Assigned is tied to one resource; User Assigned is reusable across multiple resources.

Q3. What is RBAC?

Role-Based Access Control for managing permissions.

Q4. What is APIM?

Azure API gateway for security, routing, throttling, monitoring, and transformations.

Q5. What is Application Insights?

Application performance monitoring service.

Q6. What is Azure Monitor?

Platform-wide monitoring, logging, metrics, and alerts.

Q7. What is Azure Front Door?

Global Layer 7 load balancer with WAF and intelligent routing.

Q8. What is Deployment Slot?

A deployment environment used for testing and zero-downtime releases.

Q9. Best authentication method for Azure Storage?

Managed Identity with Azure RBAC.

Q10. Why use Private Endpoints?

To access Azure services privately over a virtual network.

Q11. Difference between Azure Service Bus and Event Grid?

Service Bus is for reliable enterprise messaging; Event Grid is for event routing.

Q12. Difference between Event Hub and Service Bus?

Event Hub is for high-throughput streaming; Service Bus is for business messaging.

Q13. What is Key Vault?

Secure storage for secrets, keys, and certificates.

Q14. What is Cosmos DB?

A globally distributed, multi-model NoSQL database.

Q15. Why choose Azure App Service over Virtual Machines?

Lower operational overhead, built-in scaling, patching, and managed infrastructure.

Module 5 Progress

You've now covered:

- Azure App Service
- Azure Functions
- Storage Accounts
- Blob Storage
- Queue Storage
- Service Bus
- Event Grid
- Event Hub

- Azure SQL
- Cosmos DB
- Key Vault
- Managed Identity
- Azure RBAC
- APIM
- Application Insights
- Azure Monitor
- Storage Authentication
- Private Endpoints
- Azure Front Door
- Deployment Slots
- Production Troubleshooting Scenarios

Next Module: Azure DevOps & CI/CD (Highly Important)

The next module will cover the most frequently asked Azure DevOps interview topics for senior .NET developers:

- Azure Repos
- Git Branching Strategies
- Pull Requests & Code Reviews
- Azure Pipelines (Classic & YAML)
- Build Pipelines
- Release Pipelines
- CI/CD Best Practices
- Deployment Strategies (Blue-Green, Canary, Rolling)
- Infrastructure as Code (Bicep, ARM, Terraform)
- Docker & Kubernetes Basics
- Real-world production deployment scenarios

This module is especially relevant for your experience with **Azure DevOps and Bicep deployments** and is frequently covered in senior technical interviews.